

PART 1. General Knowledge: **Answer in short*

1. What is the basic architecture of a typical data warehouse?
2. How many type of SCD dimension exist? How do you decide which to choose?
3. How do you increase SQL Performance of DW on Oracle? What is the effect of adding too many indexes?
4. How do you decide to create dimension or fact table?
5. How do you increase ETL Performance?
6. What is vertical and horizontal scaling? What is CAP theorem?
7. What is Conceptual/Logical/Physical design?
8. Differences between Kimball and Inmon methodologies in short?
9. What is Operational Data Store (ODS)?
10. What can be the types of data sources in data warehouse?
11. Difference between a data warehouse and a data lake in short?
12. What is the usage of the 'static' keyword in Java? (*not must*)
13. One customer goes to the bank branch and deposits €100 into own account. What amount is registered to branch cash account?

PART 2. SQL Questions:

1. Write SQL query to find all numbers that appear at least 3 times consecutively. Return the result in any order.
Query should work for N times consecutives also.

Table: Logs

Colum name	Type
ID(PK)	Number
NUM	Varchar2

Example:

Input:

ID	NUM
1	1
2	1
3	1
4	2
5	1
6	2
7	2

Output: 1 is the only number that appears consecutively for at least 3 times.

Consecutive_Nums
1

2. Write a query to find the prices of all products on 2024-07-16. Assume the price of all products before any change is 10. Product_Id + change_date together are unique index

Table: Products

Column name	Type
Product_ID	Number
As_of_Date	Date
Price	Number

Example:

Input:

Product_ID	As_of_dt	Price
1	2024-07-14	20
2	2024-07-14	50
1	2024-07-15	30
1	2024-07-16	35
2	2024-07-17	65
3	2024-07-18	20

Output:

Product_ID	Price
2	50
1	35
3	10

3. Write a solution to calculate the number of bank accounts for each salary category. The salary categories are:

"Low Salary": All the salaries strictly less than €20000.

"Average Salary": All the salaries in the inclusive range [€20000, €50000].

"High Salary": All the salaries strictly greater than €50000.

The result table must contain all three categories. If there are no accounts in a category, return 0.

Table: Accounts

Column name	Type
Account_ID	Number
Income	Number

Example:

Input:

Account_ID	Income
3	108939
2	12747
8	87709
6	91796

Output:

Category	Count
Low Salary	1
Average Salary	0
High Salary	3

4. Write SQL to find the employees who are high earners in each of the departments. A **high earner** in a department is an employee who has a salary in the **top three unique** salaries for that department.

Table: Employee

Column name	Type
ID(PK)	Number
Name	Varchar
Salary	Number
Dept_ID(FK)	Number

Table: Department

Column name	Type
ID(PK)	Number
Name	Varchar

Example:

Input:

Employee

ID	Name	Salary	Dept_id
1	Radu	8500	1
2	Mircea	8000	2
3	Ahmet	6000	2
4	Esra	9000	1
5	Daniel	6900	1
6	Silvia	8500	1
7	Omer	7000	1

Department

ID	Name
1	IT
2	Sales

Output:

Department	Employee	Salary
IT	Esra	9000
IT	Radu	8500
IT	Silvia	8500
IT	Omer	7000
Sales	Mircea	8000
Sales	Ahmet	6000

5. Write SQL query to find for each month and country, the number of transactions and their total amount, the number of approved transactions and their total amount.

Table: Transaction

Column name	Type
ID(PK)	Number
Country	Varchar
Status	Varchar
Amount	Number
Trx_date	Date

Example:

Input:

ID	Country	Status	Amount	Trx_date
121	RO	approved	1000	2023-12-18
122	RO	declined	2000	2023-12-19
123	RO	approved	2000	2024-01-01
124	DE	approved	2000	2024-01-07

Output:

month	country	trans_count	approved_count	trans_total_amount	approved_total_amount
2023-12	RO	2	1	3000	1000
2024-01	RO	1	1	2000	2000
2024-01	DE	1	1	2000	2000

PART 3. Design Question: **Answer what you understand. You can generate your test data*

Company A produces 3 products: Product P1, Product P2 and Product P3.

Every month, Company A produces QP1 units of Product P1, QP2 units of Product P2 and QP3 units of Product P3.

To support profitability analysis, management of Company A requested to establish **weekly reporting on cost performance**. Weekly reporting must include:

- Reporting on total cost analysis for Company A
- Reporting on cost for each individual Product produced by Company A

You need to **design the process by data flow management perspective in order to support accurate and timely weekly reporting**. Process description should include details on each step starting **with data collection from the data sources** and **finishing with final layer for reporting** which can be used by management to get requested reporting. **Dimension, Fact tables and Final report should be designed.**

Please include following SQL queries into your answer:

- 1) SQL query to get total weekly cost (Administrative + Production cost) for Company A,
- 2) SQL query to get total weekly cost (Administrative + Production cost) for each product produced by Company A

Feel free to use tables, screen shots, and graphs in your report. Any visualization tool (Excel, PBI, ODI, tableau, etc.) can be used for the report. You can generate your own test data for the source tables

Additional source data points to support design of your solution:

1. Cost details

Production cost per unit:	
- Product P1	- CostP1/unit
- Product P2	- CostP2/unit
- Product P3	- CostP3/unit

2. Data sources:

Table 1 PRODUCTION_DATA

Product	Quantity	Cube
P1	QP1	CubeP1
P2	QP2	CubeP2
P3	QP3	CubeP3

Table 2 ADMIN_COST

Cost_type	Amount
Administrative expenses	CostAdm (monthly cost allocated to products based on cube